KWAME NKRUMAH UNIVERSITY OF SCIENCE AND TECHNOLOGY

COLLEGE OF ENGINEERING

DEPARTMENT OF ELECTRICAL/ELECTRONICS ENGINEERING

MICROPROCESSORS PROJECT WORK REPORT

GROUP B2-2

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PROJECT NAME: AUTOMATIC STREET LIGHT

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Abstract

This report presents the design and implementation of an automatic street light system aimed at improving energy efficiency and reducing unnecessary power consumption. The system uses a Light Dependent Resistor (LDR) to detect ambient light levels and automatically switches the street lights ON at night and OFF during the day. The project demonstrates how simple electronic components can be integrated to create an intelligent and cost-effective lighting solution. The system can be further enhanced using microcontrollers and renewable energy sources such as solar power.

Contents

Chapter 1

Introduction

Street lighting plays a vital role in ensuring safety, security, and improved visibility in urban and rural environments. Conventional street lighting systems are manually controlled or operate on fixed timers, which often leads to energy wastage. This project focuses on developing an automatic system that responds to environmental light conditions, thereby optimizing energy usage.

Chapter 2

Background Study

With increasing energy demand and rising electricity costs, efficient energy management has become essential. Automatic lighting systems have been widely studied and implemented in smart cities. The use of sensors such as LDRs allows systems to respond dynamically to changes in light intensity. This project builds on these concepts to provide a simple and affordable solution.

Chapter 3

Problem Statement

Traditional street lighting systems remain ON even during daylight or when not required, leading to energy wastage and increased operational costs. There is a need for an automated system that can intelligently control street lighting based on environmental conditions.

Chapter 4

Aim and Objectives

Aim

To design and implement an automatic street lighting system that operates based on ambient light levels.

Objectives

- To develop a system that automatically switches lights ON and OFF
- To minimize energy wastage
- To design a reliable and efficient circuit
- To demonstrate the working using LEDs or a relay-controlled lamp

Chapter 5

Scope of the Project

This project is limited to the design and implementation of a prototype automatic street light system using basic electronic components. It focuses on light-based automation and does not include advanced features such as wireless monitoring or IoT integration.

Chapter 6

Methodology

The project was carried out using the following steps:

1. Research on automatic lighting systems
2. Selection of suitable components
3. Circuit design and simulation
4. Hardware implementation on breadboard
5. Testing and verification
6. Documentation of results

Chapter 7

System Design

The system is designed using an LDR as the sensing element, a transistor as a switch, and an LED or relay as the output device. A voltage divider circuit is used to provide the required biasing for the transistor.

Chapter 8

Components Description

Light Dependent Resistor (LDR)

A sensor whose resistance varies with light intensity.

Resistors

Used to limit current and form voltage divider circuits.

Transistor (BC547)

Acts as a switch to control the output device.

Relay / LED

Represents the street light in the system.

Power Supply

Provides the required voltage for operation.

Chapter 9

Working Principle

During daylight, the LDR has low resistance, resulting in low voltage at the transistor base, keeping it OFF. At night, the resistance of the LDR increases, raising the voltage at the base of the transistor, which turns it ON and activates the light.

Chapter 10

Implementation

The circuit was assembled on a breadboard. Proper connections were made according to the design. The LDR was positioned to detect ambient light, and the LED was used to indicate the operation of the system.

Chapter 11

Testing and Results

The system was tested under different lighting conditions:

- In bright light: LED remained OFF
- In darkness: LED turned ON

The system responded accurately to changes in light intensity, confirming its proper functionality.

Chapter 12

Applications

- Street lighting systems
- Garden lighting
- Parking areas
- Security lighting
- Highway illumination

Chapter 13

Personal Contribution

In the course of this project, I was actively involved in the design, construction, and testing of the automatic street light system. I carried out component selection, assembled the circuit, and ensured proper functioning through testing. Additionally, I contributed to troubleshooting errors and improving the overall performance of the system.

Chapter 14

Conclusion

The automatic street light system developed in this project successfully demonstrates how energy can be conserved using simple electronic components. The system is reliable, efficient, and cost-effective, making it suitable for real-world applications.

Chapter 15

Recommendations

- Integration with solar power systems
- Use of motion sensors for enhanced efficiency
- Implementation using microcontrollers for smart control

Chapter 16

References

- Basic Electronics Textbooks
- Online Engineering Resources
- Arduino Documentation